

Dr. Lenard Dome

Learning, generalisation, and model adequacy
University of Tübingen
German Center for Mental Health (DZPG)

✉ lenard.dome@uni-tuebingen.de
🌐 <https://lenarddome.com/>
🐱 [lenarddome](#)
🐦 [lenarddome](#)

RESEARCH STATEMENT

I work on theories of learning and cognition – learning dynamics, generalisation, and the formal evaluation of network models. I study how representations emerge, reorganise, and generalise under different training regimes, and how to tell whether a model reproduces a behaviour for the right reasons. My focus, across network models of human learning, is the conditions under which it fails rather than succeeds: failure modes constrain the space of admissible mechanisms far more sharply than successes do, and those derived in small, tractable models are worth extrapolating to larger systems – a derivation tells you where to look, not simply what happened.

POSITIONS AND EDUCATION

2023 - Postdoctoral researcher. University of Tuebingen. (Advisor: Tobias Hauser)

2023 - Guest Scientist. Max Planck Institute for Biological Cybernetics.

2021 - 2023 Visiting Lecturer, Tutor and Specialist. University of Plymouth.

2020 - 2023 Ph.D. Psychology, Computational Cognitive Science. University of Plymouth. *No corrections.*
(Supervisor: Andy Wills)

- Thesis: *Testing the adequacy of formal models of an irrational learning effect.*

2019 - 2020 M.Sc. Advanced Psychology. University of Plymouth. *Distinction.* (Supervisor: Patric Bach)

2017 - 2019 B.Sc. Psychology. University of Plymouth. *First.*

2017 - 2019 Research Assistant. University of Plymouth. (Line managers: Andy Wills, Marina Wimmer)

CURRENT RESEARCH PROJECTS

Stabilising attentional learning across training regimes (*planned*)

- Work on network models of attentional learning, aimed at building and improving architectures to remain stable (develop robust representations in tractable models) across supervised, unsupervised, associative and reinforcement learning, and across increasing task complexity. It extends my work (Dome, 2026) showing that shared attention learned through gradient-descent collapses under multi-outcome learning, which outcome-indexed attention resolves.

Representational reorganisation under intervention (*early stage*)

- Simulation-based inference with attention-based networks, asking how direct interventions, analogous to exposure therapy, reorganise learned representations. Preliminary: reorganisation occurs only within the input space the attentional tuning already covers, and newly learned connections are overshadowed during generalisation – an intervention targeting avoidance fails for the same reason avoidance persists, because it confines the representation with narrow attentional tuning.

PUBLICATIONS

2026 **Dome, L.** “Why Shared Attention Vectors Fail: A Case for Outcome-Indexed Tuning.” *arXiv*. [10.48550/arXiv.2609.08615](https://arxiv.org/abs/10.48550/arXiv.2609.08615) (under review at *Biological Cybernetics*)

- Derives when gradient-learned shared attention collapses under multi-outcome learning, and introduces an outcome-indexed fix, validated in three synthetic experiments.

- 2026 **Dome, L.**, Wills, A.J. “Errorless Irrationality: A unified computational account of the inverse base-rate effect across predictive, observational, and unsupervised procedures.” *arXiv*. [10.48550/arXiv.2608.06149](https://arxiv.org/abs/10.48550/arXiv.2608.06149)
- One auto-associative network reproduces the same generalisation failure under supervised, observational and unsupervised training: it emerges from pattern completion, not error-driven updating, and survives removing the error signal entirely.
- 2026 **Dome, L.**, Moses-Payne, M. E., Skvortsova, V., & Hauser, T. U., “Age-related shifts in metacognition reveal converging confidence for correct and error judgments across the lifespan.” *PsyArXiv*. [10.31234/osf.io/ugb9s](https://doi.org/10.31234/osf.io/ugb9s)
- Signal Detection Theoretic modelling of how confidence estimates decouple from accuracy, in a 6K+ lifespan sample.
- 2026 **Dome, L.**, Hezemans, F. H., Kadri, K., Wagner, B. J., Webb, A. & Hauser, T. U. “cpm: a python library for theory-driven modeling in Computational Psychiatry.” *PLOS Computational Biology*. [10.1371/journal.pcbi.1014481](https://doi.org/10.1371/journal.pcbi.1014481).
- A Python library for specifying, fitting, comparing and recovering theory-driven models.
- 2026 Schlegelmilch, R. & **Dome, L.** Snake oil or Panacea? How to Misuse AI in Scientific Inquiries of the Human Mind. *Behavioral Sciences*. [10.3390/bs16020219](https://doi.org/10.3390/bs16020219).
- On the failure modes of substituting black-box models for mechanistic theory in scientific inference about the mind.
- 2026 Marzuki, A., Kosina, L., **Dome, L.**, Hewitt, S., & Hauser, T. Drops in Confidence Precede Symptoms of Obsessive-Compulsive Disorder. *Nature Communications*. [10.1038/s41467-026-76671-5](https://doi.org/10.1038/s41467-026-76671-5).
- Longitudinal evidence that a shift in a latent internal variable precedes the onset of OCD symptoms.
- 2025 **Dome, L.** & Wills, A.J. “g-Distance: On the comparison of model and human heterogeneity.” *Psychological Review*. [10.1037/rev0000550](https://doi.org/10.1037/rev0000550)
- Compares whether a model’s spread of behaviours matches the spread in human data, not just the average.
- 2025 **Dome, L.** & Wills, A.J. “Better generalization through distraction? Concurrent load reduces the size of the inverse base-rate effect.” *Psychonomic Bulletin & Review*. [10.3758/s13423-025-02661-1](https://doi.org/10.3758/s13423-025-02661-1)
- An experimental test of a prediction derived from the model above: restricting available capacity shrinks the generalisation failure.
- 2023 **Dome, L.** & Wills, A. J. “Errorless irrationality: removing error-driven components from the inverse base-rate effect paradigm.” *Proceedings of the 45th Annual Conference of the Cognitive Science Society*. Austin, TX: Cognitive Science Society. escholarship.org/uc/item/0kw671vv
- 2021 **Dome, L.**, Edmunds, C. E. R., & Wills, A. J. “SUSTAIN captures category learning, recognition, and hippocampal activation in a unidimensional vs information-integration task.” *Proceedings of the 43rd Annual Conference of the Cognitive Science Society*. Austin, TX: Cognitive Science Society. escholarship.org/uc/item/5r98q3dr
- 2021 Spicer, S. G., Wills, A. J., Jones, P. M., Mitchell, C., & **Dome, L.** “Representing uncertainty in the Rescorla-Wagner model: blocking, the redundancy effect, and outcome base rate.” *Open Journal of Experimental Psychology and Neuroscience*. doi.org/10.46221/ojepn.2021.6623
- 2019 Wimmer, M. C., **Dome, L.**, Wennekers, T. & Hancock, P. J. B. “Is the letter cancellation task a suitable index of ego-depletion? Empirical and conceptual issues.” *Social Psychology*. doi.org/10.1027/1864-9335/a000393

RESEARCH SOFTWARE

-  **cpm** *maintainer, developer* A Python library for theory-driven modelling: model specification, fitting, comparison, and recovery. Published in *PLOS Computational Biology*; adopted by modelling labs internationally. [\[source code\]](#)
-  **psp** *maintainer, developer* An n -dimensional parameter space partitioning algorithm (after Pitt, Kim, Navarro & Myung, 2006) that identifies the boundaries in parameter space corresponding to each discrete behavioural outcome a model can produce, rather than fitting a point estimate. 18K downloads. [\[source code\]](#) [\[cran\]](#) doi: [10.32614/CRAN.package.psp](https://doi.org/10.32614/CRAN.package.psp)
-  **catlearn** *senior developer* An archive of formal models of learning and categorization, plus benchmark datasets to test them against. It currently houses 14 models; I have contributed 8. 57K downloads. [\[source code\]](#) [\[cran\]](#) doi: [10.32614/CRAN.package.catlearn](https://doi.org/10.32614/CRAN.package.catlearn)

COMPUTATION

Python (NumPy, SciPy, pandas, PyTorch, scikit-learn), C++, R, Bash, Git, Docker, Synology; large-scale parameter-space simulation and model recovery on HPC/SLURM clusters.

Certificate in High-Performance Computing with OpenMP & MPI, HLRS, University of Stuttgart (2024).

GRANTS AND AWARDS

- 2026 **Awarded**
DZPG Travel Grant.
- 2020 **Awarded**
[Undergraduate Project Prize](#). Experimental Psychological Society / British Science Association.
- 2019 **Awarded**
Doctoral Training Studentship (1+3). ESRC & SWDTP. £87,510.62.

INVITED TALKS

Yale University, USA (MAP Talk, 2026); University of Tübingen, Germany (Psychiatry Colloquium, 2024); British Science Association / Experimental Psychological Society, UK ([Undergraduate Project Prize Talk](#), 2021).

CONFERENCE PRESENTATIONS

Research Colloquium, Tübingen (2026); DZPG Early Career Congress, Berlin (2026); Computational Psychiatry Conference, Tübingen (2025); Cognitive Science Society, Sydney (2023) and online (2021); Math-Psych/ICCM/EMPG, Amsterdam (2023); Experimental Psychology Society, Plymouth (2023); Associative Learning Symposium, Gregynog Hall, Wales (2022).

TEACHING, SUPERVISION AND SERVICE

Supervised [CaCTüS](#) internship students at the Max Planck Institute for Biological Cybernetics, Germany (2025–2026) and final-year B.Sc. projects at the University of Plymouth, UK (2021–2023). Taught computational modelling from B.Sc. to postdoctoral level at the University of Tübingen, Germany (2024–2026), and research methods, quantitative analysis and data visualization in R at Plymouth (B.Sc., DCLinPsy, 2021–2023). Reviewer for *The American Statistician*.